

AI MEDICAL WASTE CLASSIFIER

Multi-Step Vision & Prompt Workflow for Biohazard Sorting & Tracking

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ASSESSMENT: Mini Assessment — Prompt Workflow Design	

01 | Concept

Purpose: The AI Medical Waste Classifier processes images of hospital waste at disposal stations, classifies the items into precise regulatory categories (e.g., Biohazard, Sharps, General), retrieves local safety compliance guidelines, generates tracking manifests, and flags hazardous mixing for immediate human intervention.

Problem: Improper disposal of medical waste leads to severe environmental hazards and hefty compliance fines. A multi-step multimodal AI pipeline standardizes waste evaluation through deep learning image processing, ensuring safe routing and automated compliance tracking.

02 | Objective

The workflow uses sequential prompts and vision models to analyze visual waste data, categorize risk levels, query hospital safety protocols, auto-generate tracking logs, and evaluate classification confidence before finalizing disposal routes.

03 | Workflow Overview

1. Scan	2. Classify	3. Retrieve	4. Draft Logs	5. Audit
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04 | Workflow Steps

- 1. Image Parsing & Feature Extraction: Analyze visual feed from disposal bins to identify items (e.g., syringes, gloves, paper). [Output: Structured Item Data]
- 2. Risk Classification: Categorize waste into Red Bag (Biohazard), Sharps, Trace Chemotherapy, or General Refuse. [Output: Risk Category + Disposal Route]
- 3. Regulation Matching: Retrieve pertinent environmental regulations and handling procedures from the hospital knowledge base. [Output: Compliance Context]
- 4. Manifest Generation: Draft a standardized tracking log entry including timestamp, location, and identified materials. [Output: Draft Tracking Manifest]

- 5. Escalation & Quality Check: Verify AI confidence scores. If confidence is below 95% or conflicting items (e.g., paper mixed with sharps) are detected, lock bin and alert staff. [Output: Final Action / Escalation Alert]

05 | Prompt Workflow

Step 1 — Parse Image Feed

Analyze the incoming image. Identify all distinct objects in the frame, their material composition (plastic, glass, organic), and note any visible fluids or contamination.

Step 2 — Classify Risk Level

Based on the extracted objects, assign the strictest applicable risk category (Sharps > Biohazard > General) and determine the designated disposal bin color code.

Step 3 — Retrieve Compliance Docs

Match the identified risk category against the hospital waste management policy. Select the top compliance rules for handling and maximum storage time.

Step 4 — Draft Tracking Manifest

Using the classified data and compliance rules, format a JSON tracking manifest containing the station ID, identified objects, hazard level, and required pickup SLA.

Step 5 — Confidence & Safety Audit

Review the classification confidence. If the AI detects a mixed-waste violation (e.g., a needle in the general waste stream) or confidence is < 0.95, override auto-sorting and trigger a human review alert.

06 | Prompt Chaining Architecture

Step 1 produces visual metadata; Step 2 determines the hazard route; Step 3 gathers regulatory context; Step 4 crafts the digital trail; and Step 5 audits safety thresholds.

STAGE	INPUT / CONTEXT	OUTPUT
1	Raw Camera Image	Extracted Object List & Materials
2	Step 1 Object List	Risk Category & Bin Assignment
3	Step 2 Category + KB Docs	Handling Guidelines & SLA
4	Step 3 Rules + Step 1 Details	JSON Tracking Manifest
5	Step 4 Draft + Confidence Score	Approved Log / Supervisor Alert

07 | Demonstration Input

Example Scenario: A disposal station camera captures an image showing a used syringe resting on top of standard plastic packaging wrappers, initially tossed toward the "General Waste" chute.

08 | Expected Final Output & Workflow Result

An automated risk classification of "Critical / Sharps Violation", retrieved mixed-waste penalty protocols, an aborted sorting action, and an immediate alert sent to the floor supervisor.

10 | Sample Support Execution Plan

PHASE	ACTION FOCUS	OUTPUT / METRIC
Phase 1	Object Detection	Syringe detected → Vision model flags sharp object
Phase 2	Risk Assessment	Mixed waste confirmed → Sharps protocol initiated
Phase 3	Log Drafting	Metadata captured → Manifest prepared with violation tag
Phase 4	Safety Evaluation	Violation threshold met → Routed to Floor Supervisor
Phase 5	Resolution Track	Bin locked → Human agent clears violation within 2 mins

11 | Key Benefits

- ✓ Eradicates human error in high-risk biohazard sorting
- ✓ Automated compliance with environmental and healthcare regulations
- ✓ Real-time tracking of waste volume and hazard distribution
- ✓ Immediate flagging of cross-contamination events
- ✓ Highly scalable architecture utilizing vision-language models (VLMs)

12 | Discussion Questions & Answers

1. What is the advantage of using {user_input} instead of hard-coding input?

Templating separates the prompt's fixed structure (e.g., instructions for classifying waste) from the variable content (the specific image features or object list). This allows the medical waste classifier to scale and handle thousands of unique disposal events dynamically without hard-coding rules for every single interaction.

2. When would .format() be enough?

Python's .format() is sufficient for simple, linear prompt templates where variables (like a list of detected items) just need to be directly inserted into predefined slots. It is ideal when the prompt does not require complex conditional logic or nested instructions.

3. When might Jinja2 be useful?

Jinja2 is highly useful for the waste classifier when conditional logic is needed. For example, using if/else blocks to append specific hazardous material handling instructions only if the vision model detects 'Sharps' or 'Biohazard' items, keeping the prompt concise for standard general waste.

4. Why should prompts be versioned?

Versioning prompts ensures reproducibility and provides an audit trail. In a strict healthcare compliance setting, if a new prompt version inadvertently causes biohazard misclassification, versioning allows for an immediate, safe rollback to a known working state.

5. What is a prompt regression?

A regression occurs when a prompt adjustment (e.g., tweaking the system prompt to better identify glass vials) unintentionally degrades the model's performance on previously mastered tasks (like identifying plastic syringes). It highlights the interconnected nature of LLM instructions.

6. Why use the same evaluation set when comparing prompt versions?

Using a static evaluation set (a 'golden dataset' of standard and edge-case waste images) ensures an apples-to-apples comparison. It guarantees that any change in classification accuracy is strictly due to the prompt edit, rather than variations in the test data's difficulty.